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Covered papers

Paper 1

- ☐ 2025 Winter 13
- ☐ 2025 Winter 12
- ☐ 2025 Winter 11

- ☐ 2025 Summer 13
- ☐ 2025 Summer 12
- ☐ 2025 Summer 11
- ☐ 2025 March

Command words

Command words and their meanings help candidates know what is expected from them in the exam. The table below includes command words used in the assessment for this syllabus. The use of the command word will relate to the subject context.

Analyse	examine in detail to show meaning, identify elements and the relationship between them
Compare	identify/comment on similarities and/or differences
Contrast	identify/comment on differences
Define	give precise meaning
Demonstrate	show how or give an example
Describe	state the points of a topic/give characteristics and main features
Discuss	write about issue(s) or topic(s) in depth in a structured way
Evaluate	judge or calculate the quality, importance, amount, or value of something
Explain	set out purposes or reasons/make the relationships between things clear/say why and/or how and support with relevant evidence
Give	produce an answer from a given source or recall/memory
Identify	name/select/recognise
Justify	support a case with evidence/argument
State	express in clear terms
Suggest	apply knowledge and understanding to situations where there are a range of valid responses in order to make proposals/put forward considerations

1 Types and components of computer systems

1.1 Hardware and software

W25_QP11_Q2

Data can be in two forms, analogue and digital.

(a) Complete the following sentences about the different types of data.

(i) The data read by a sensor is

.....[1]

(ii) The data read by a microprocessor is

.....[1]

(b) **Contrast** analogue data with digital data.

.....
.....
.....
.....
.....
.....
.....
.....[4]

S25_QP13_Q2

A computer system uses both applications software and system software.

(a) Audio editing software is an example of applications software.

Describe what is meant by audio editing software.

.....
.....
.....
.....[2]

(b) **State** two examples of system software.

.....
.....

.....
.....[2]

S25_QP12_Q2

A computer system uses both applications software and system software.

(a) Describe the following types of applications software.

(i) Computer aided design (CAD)

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.....
.....
.....[2]

(ii) Measurement software

.....
.....
.....
.....[2]

(iii) Video editing software

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.....
.....
.....[2]

(b) State the purpose of applications software.

.....
.....[1]

S25_QP11_Q2

A computer system processes data by using applications software and system software.

(a) **Describe** the following types of applications software.

(i) Word processing

.....

.....

.....

.....[2]

(ii) Control software

.....

.....

.....

.....[2]

(iii) Apps

.....

.....

.....

.....[2]

(b) Data can be in two forms, analogue or digital.

Explain why analogue data needs to be converted to digital data.

.....

.....[1]

(c) **Explain** why digital data needs to be converted to analogue data.

.....

.....[1]

M25_QP12_Q2

A computer system uses applications software and system software.

(a) **Describe** the following software.

(i) Spreadsheet

.....
.....
.....
.....[2]

(ii) Database

.....
.....
.....
.....[2]

(iii) Applet

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.....
.....
.....[2]

(b) **State** three types of system software.

.....
.....
.....
.....
.....
.....[3]

1.2 The main components of computer systems

W25_QP11_Q1

Circle **two** examples of internal memory

Actuator	CD-ROM	Hard drive	Mouse	
Pen drive	RAM	ROM	SD card	[2]

W25_QP11_Q5

Instructions are entered into a computer and processed to produce an output.

Explain the role of the CPU in processing the instructions.

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.....

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.....

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.....[4]

1.3 Operating systems

1.4 Types of computer

W25_QP11_Q6

A school is planning to create a computer suite in its library. The headteacher is considering using desktop computers or mobile computers.

(a) Discuss the benefits and drawbacks of the headteacher choosing desktop computers.

[6]

1.5 Emerging technologies

W25_QP11_Q7

Tawara is an online shopping company.

(b) Discuss the impact of using Artificial Intelligence (AI) in online shopping.

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.....

.....

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.....[4]

2 Input and output devices

2.1 Input devices and their uses

M25_QP12_Q1

Circle two input devices.

Actuator	Blu-ray disc	Cloud	Keyboard
Flash Memory	Hard disk	Mouse	SD card

[2]

2.2 Direct data entry and associated devices

S25_QP12_Q1

Circle two direct data entry devices.

3D printer	DVD reader	keyboard	mouse	
Pen drive	QR scanner	RFID reader	SD card	[2]

2.3 Output devices and their uses

W25_QP11_Q4

A design company is planning to buy either a laser printer or a plotter.

Compare and contrast a laser printer with a plotter.

[6]

S25_QP13_Q3

An architect is planning to buy a plotter.

Discuss the benefits and drawbacks of the architect using a plotter.

[6]

S25_QP11_Q1

Circle two output devices.

Actuator

Blu-ray disc

Cloud

Keyboard

Hard disk

Printer

SD card

Webcam

[2]

3 Storage devices and media

3.1 Storage devices and media

W25_QP11_Q9

(b) State one example of each of the following types of storage media.

(i) optical

.....[1]

(ii) solid-state

.....[1]

S25_QP13_Q1

Circle **two** optical storage devices.

Blu-ray disc reader

DVD reader

Keyboard

Mouse

Pen drive

QR scanner

RFID reader

SD card

[2]

M25_QP12_Q8

State one example of a device that can be used to read data from the following storage media.

(a) Optical

.....[1]

(b) Magnetic

.....[1]

(c) Solid State

.....[1]

4 Networks and the effects of using them

4.1 Networks

W25_QP11_Q8

A company uses the cloud to store data and software.

(a) Explain what is meant by the term cloud.

.....[3]

(b) Access to the cloud is via the internet. **Describe** the characteristics of the internet.

[6]

S25_QP13_Q7

Tawara Cars uses a private cloud to store data and software.

(a) Explain what is meant by a private cloud.

.....

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.....[3]

- (b) Some of the programmers at Tawara Cars are working together on creating a new design package which is stored on the cloud.

Describe how data can be stored and managed using cloud computing

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.....[4]

S25_QP12_Q9

A company's computer network uses a switch.

- (a) **Explain** the purpose of a network switch.

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.....
.....
.....[3]

- (b) A smart TV can connect to a network using wi-fi.

Describe how the smart TV can be connected to the wi-fi.

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.....
.....
.....[4]

(c) **State** the type of network that the smart TV would connect to

.....[1]

S25_QP12_Q11

A student is working on a project and needs to transfer some of her files from the college computer to her home computer. One way in which the student could transfer the files is uploading them onto the cloud.

(a) **State** three other ways in which the student can transfer the files.

.....
.....
.....
.....
.....
.....[3]

(b) **Describe** how the student could use the cloud to transfer the project.

.....
.....
.....
.....
.....
.....[3]

S25_QP11_Q9

A local area network (LAN) uses a hub.

- (a) Explain the purpose of a hub**

.....[3]

- (b) Describe** one drawback of using a hub on a local area network (LAN).

.....[1]

M25_QP12_Q5

The cloud can be used for computer gaming. One major issue is the protection of personal data.

- (a) Discuss**, apart from the protection of personal data, the benefits and drawbacks of gaming on the cloud.

[6]

M25_QP12_Q9

Computers use Network Interface Cards (NIC).

- (a) Explain** the purpose of a Network Interface Card (NIC).

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.....[4]

(b) A user wishes to connect a set of Bluetooth headphones to a computer.

Describe how the headphones can be connected to the computer by Bluetooth.

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.....

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.....[4]

4.2 Network issues and communication

W25_QP11_Q9

An employee in an organisation has been working at home and has saved a file on his portable hard disk drive (HDD). He takes the portable HDD into the office.

- (a) **Describe** the process of safely uploading the file onto the organisation's network. Include in your answer any issues that may occur and how they can be avoided.

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.....

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.....[4]

5 The effects of using IT

5.1 Microprocessor-controlled devices

5.2 Potential health problems related to the prolonged use of IT equipment

W25_QP11_Q7

(c) The use of computer equipment can lead to many different health issues.

For each of the following health issues, **describe** three different ways of minimising the risk.

(i) eyestrain

.....

.....

.....

.....

.....

.....[3]

(ii) back problems

.....

.....

.....

.....

.....

.....[3]

S25_QP13_Q6

The use of computer equipment can lead to many different health issues.

For each of the following health issues **describe** three different ways of minimising the risk.
Your answers must be different in each case.

(a) Repetitive Strain Injury (RSI)

.....

.....

.....

.....

.....

.....[3]

(b) Headaches

.....
.....
.....
.....
.....
.....[3]

6 ICT applications

6.1 Communication

S25_QP11_Q3

Evaluate the use of smartphones for making video calls.

[6]

S25_QP11_Q5

Viewers can watch movies by video streaming. One of the issues with video streaming can be a reduction in quality due to file compression.

(a) Discuss the use of video streaming to watch movies

[6]

6.2 Modelling applications

S25_QP13_Q8

Weather forecasts are generated using computer models.

Discuss the benefits and drawbacks of using computer models in this way.

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.....[5]

6.3 Computer controlled systems

6.4 School management systems

W25_QP11_Q6

A school is planning to create a computer suite in its library. The headteacher is considering using desktop computers or mobile computers.

- (b) The headteacher is planning to use the computers for computer aided learning (CAL).

Explain what is meant by computer aided learning (CAL).

.....

.....

.....

.....

.....

.....[3]

- (c) **Describe** three features of CAL.

.....

.....

.....

.....

.....

.....[3]

6.5 Booking systems

6.6 Banking applications

S25_QP12_Q12

State the processing involved in withdrawing cash from an Automatic Teller Machine (ATM) after a customer has inserted his card into the machine.

.....

.....

.....

.....

.....

.....[3]

6.7 Computers in medicine

6.8 Expert systems

M25_QP12_Q3

A doctor uses an expert system to diagnose a patient's illness.

(a) **Describe** how expert systems produce diagnoses of an illness.

.....

.....

.....

.....

.....

.....

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.....[4]

(b) **Explain** how an expert system could produce an incorrect diagnosis of an illness.

.....

.....

.....

.....[2]

6.9 Computers in the retail industry

W25_QP11_Q7

Tawara is an online shopping company.

(a) Describe the characteristics of online shopping.

[6]

6.10 Recognition systems

S25_QP12_Q3

The headteacher at Tawara School is planning to improve the registration system for pupils. The headteacher can choose a system that uses either ID cards with a magnetic stripe or scanners that read handprints.

Discuss the benefits and drawbacks of using a system that will read handprints.

[6]

6.11 Satellite systems

S25_QP12_Q7

A group of people use satellite navigation systems on their smartphones when walking in remote areas of the countryside.

Discuss the benefits and drawbacks of using satellite navigation systems in this scenario.

[6]

S25_QP11_Q6

Many taxis now use satellite navigation systems.

Evaluate the use of satellite navigation systems for taxis.

[6]

M25_QP12_Q6

Some friends are planning to explore a remote area of India. They need to stay in contact with their families and can either use a smartphone or a satellite phone.

Evaluate the use of a satellite phone rather than a smartphone in this scenario.

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.....[6]

7 The systems life cycle

7.1 Analysis

S25_QP11_Q10

During the analysis of a current system, research needs to be carried out.

(a) **Describe** the characteristics of each of the following research methods

(i) Observation

.....
.....
.....
.....[2]

(ii) Interviews

.....
.....
.....
.....[2]

M25_QP12_Q10

A company is planning to replace its computer system. It will need to carry out an analysis of the current system.

(a) **State** the information about the current system that needs to be identified.

.....
.....
.....
.....
.....
.....[3]

(b) A system specification is created during the analysis.

Explain the purpose of a system specification.

.....
.....
.....

.....
.....
.....[3]

7.2 Design

W25_QP11_Q3

A headteacher has decided that all pupil reports are to be typed into a computer. He has asked the computer technician to design a layout for a pupil report.

- (a) Describe** five features of a well-designed report layout.

[5]

- (b)** The report will be linked to a database. One of the fields in the database is date of birth. An example of the data found in this field is 30/01/2010.

State two validation checks that could be used to validate the data in this field.

.....[2]

S25_QP13_Q9

A programmer has created a database for Tawara International Stores. One of the fields is Country_name which stores data as a three-letter code. For example, NZL is the code for New Zealand.

- (a) **Describe** three benefits of using these codes.

.....[3]

S25_QP12_Q4

A computer system needs to be tested before it can be implemented.

(a) **State** the purpose of applications software.

(i) File structure

.....
.....[1]

(ii) Validation routines

.....
.....[1]

(iii) The output format

.....
.....[1]

S25_QP12_Q8

A bookshop is setting up a database of the books that it sells. Part of the database is shown.

Name_of_book	Type_of_book	Price	Date_of_publication
Baines Brief Lives	N	12.99	23/03/2023
Spice	F	24.99	06/03/2022

(a) For each of the validation checks, **state** the most appropriate field. Your answers must be different in each case.

(i) Format check

.....[1]

(ii) Range check

.....[1]

(iii) Length check

.....[1]

(b) Validation and verification checks are carried out on the data.

Explain why both validation and verification are needed.

.....

[4]

(c) **State** one method of verification.

.....
[1]

(b) Following the analysis, design of the new system will take place.

Identify four items that need to be produced as part of the design of the new system.

.....

.....

.....

.....[4]

7.3 Development and testing

W25_QP11_Q3

A headteacher has decided that all pupil reports are to be typed into a computer. He has asked the computer technician to design a layout for a pupil report.

- (c) The report will be linked to a database. One of the fields in the database is date of birth. An example of the data found in this field is 30/01/2010.

One way of testing the data in this field is to use different types of test data. The first piece of test data used is acceptable data.

State which type of test data is acceptable.

.....[1]

S25_QP13_Q4

A software developer is producing a new app. The app will need to be tested before it is implemented

- (a) **State** why the following elements need to be tested.

- (i) Data structure

.....
.....[1]

- (ii) Input format

.....
.....[1]

- (b) The computer app needs to be tested with normal, extreme and abnormal data before it can be implemented.

Explain what is meant by normal test data.

.....
.....[1]

- (c) The computer app allows inputs of examination grades from grade A to grade G.

State one example of extreme data used to test the input of the examination grades

.....
.....[1]

S25_QP12_Q4

A computer system needs to be tested before it can be implemented.

(b) Computer software can be tested using normal data, extreme data or abnormal data.

Explain what is meant by abnormal test data.

.....
.....[1]

S25_QP11_Q4

Computer software needs to be tested with normal, extreme and abnormal data before it can be implemented.

(a) **Explain** what is meant by extreme test data.

.....
.....[1]

(b) An input into the software is an integer between 1 and 10. The software uses a range check to test this value as it is entered.

State one piece of test data, from each of the following, that would be used to test the range check is working correctly.

(i) Normal test data

.....[1]

(ii) Extreme test data

.....[1]

(iii) Abnormal test data

.....[1]

S25_QP11_Q10

(c) Following the implementation of the new system it needs to be tested.

Explain why the new system needs to be tested.

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.....
.....
.....
.....
.....[3]

(d) A test plan includes expected outcomes of a test.

Write down three other elements found in a test plan.

.....
.....
.....[3]

M25_QP12_Q4

A company is creating a customer database. The database needs to be tested with live data.

(a) **Explain** what is meant by live data.

.....
.....[1]

(b) **Explain** the issues associated with using live data to test this database.

.....
.....
.....
.....[2]

(c) A test strategy was produced for this database

Explain what is meant by a test strategy.

.....
.....
.....
.....[2]

7.4 Implementation

S25_QP11_Q8

State the method of implementation from the following descriptions.

- (a) The old system and the new system run together until the whole of the new system is implemented.

.....[1]

- (b) The new system is implemented one branch at a time.

.....[1]

- (c) The new system is implemented one process at a time.

.....[1]

M25_QP12_Q7

A company is planning to implement a new computer software system. It is going to use direct changeover. One of the reasons for using this method is that the benefits are immediate.

- (a) **Discuss** other benefits and drawbacks of using this method of implementation.

.....
.....
.....
.....
.....
.....
.....
.....[4]

- (b) **Explain** what is meant by the benefits are immediate.

.....
.....
.....
.....[2]

7.5 Documentation

S25_QP13_Q5

Following the development and testing of a new computer program, user documentation is produced.

(a) **Explain** why user documentation is needed.

.....
.....[1]

(b) The input format is one component in user documentation.

State three other components found in user documentation.

.....
.....
.....
.....
.....
.....[3]

(c) User documentation can be produced as an e-publication rather than a printed booklet.

Describe four advantages of producing user documentation in this format rather than a printed booklet.

.....
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.....
.....
.....
.....
.....
.....[4]

S25_QP12_Q4

- (c) Following the development and testing of the new system, technical documentation is produced.

Explain why technical documentation is needed.

.....

.....

.....

.....[2]

- (d) The file structure, validation routines and output formats are all components of technical documentation.

State three other components found in technical documentation.

.....

.....

.....

.....

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.....[3]

7.6 Evaluation

S25_QP13_Q9

A programmer has created a database for Tawara International Stores. One of the fields is Country_name which stores data as a three-letter code. For example, NZL is the code for New Zealand.

- (b) The database needs to be evaluated. Part of the evaluation is to identify any limitations.

Describe four other tasks carried out during the evaluation.

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.....

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.....

.....

.....

.....

.....[4]

8 Safety and security

8.1 Physical safety

S25_QP12_Q5

The use of computer equipment can lead to many different safety issues.

For each of the following safety issues, **describe** three different ways of minimising the risk.

(a) Electrocution

.....

.....

.....

.....

.....

.....[3]

(b) Trailing leads

.....

.....

.....

.....

.....

.....[3]

8.2 eSafety

W25_QP11_Q9

An employee in an organisation has been working at home and has saved a file on his portable hard disk drive (HDD). He takes the portable HDD into the office.

- (c) Data within the organisation needs to be protected. The organisation follows data protection legislation. **Explain** why data protection legislation is required.

.....

.....

.....

.....[2]

S25_QP13_Q10

Some personal data can be sensitive data

- (a) **Explain** the difference between personal data and sensitive data.

.....

.....

.....

.....[2]

- (b) State two examples of personal data.

.....

.....

.....

.....[2]

- (c) **State** two examples of sensitive data

.....

.....

.....

.....[2]

S25_QP12_Q10

- (c)** A search engine could be used to find the bank's website.

Describe the benefits and drawbacks of using a search engine to find the website.

[5]

M25_QP12_Q12

Parents are worried about their children's use of the internet.

- (b) Explain** how the parents could control the use of the internet by their children.

.....[3]

8.3 Security of data

W25_QP11_Q10

Phishing can be a threat to the data stored on computer systems.

(a) Describe what is meant by the term phishing.

.....[4]

(b) State three ways in which a user can attempt to prevent phishing attacks.

.....[3]

S25_QP12_Q6

(a) **Compare and contrast** phishing and pharming

[6]

(b)

- (i) **State** two precautions that the user should take to protect their data from phishing attacks.

.....

.....

.....

.....[2]

- (ii) **State** two precautions that the user should take to protect their data from pharming attacks.

.....

.....

.....

.....[2]

S25_QP12_Q10

A bank's website uses secure socket layer (SSL).

(a) **Explain** what is meant by SSL.

.....

.....

.....

.....

.....

.....[3]

(b) **State** how a user would know that SSL was being used on the website.

.....

.....[1]

S25_QP11_Q13

(a) **Describe** what is meant by hacking.

.....

.....

.....

.....[2]

(b) One method of preventing a system being hacked is the use of passwords.

Describe the password rules that should be put in place to reduce the threat of hacking.

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.....

.....[3]

M25_QP12_Q5

(b) Encryption can be used to protect personal data when gaming.

Explain what is meant by encryption.

.....

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.....[2]

9 Audience

9.1 Audience appreciation

M25_QP12_Q11

A company is developing a piece of software. The company needs to have a clear sense of audience for its software so that it will be suitable for the users.

One of the characteristics that needs to be considered is the expectations of the user.

(a) **State** two other characteristics that need to be considered.

.....
.....[2]

9.2 Copyright

M25_QP12_Q11

A company is developing a piece of software. The company needs to have a clear sense of audience for its software so that it will be suitable for the users.

- (b)** The developers are concerned about their software being pirated.

Describe three methods the company could use to help prevent software piracy.

[6]

10 Communication

10.1 Communication with other ICT users using email

S25_QP11_Q12

When sending emails users should follow netiquette.

- (a) **Explain** what is meant by netiquette.

.....[2]

- (b)** One of the rules of netiquette is to avoid sending the same message repeatedly as this is considered to be spam.

Describe, giving an example, **three** other rules used in netiquette.

.....[6]

M25_QP12_Q5

The cloud can be used for computer gaming. One major issue is the protection of personal data.

- (c) One player wants to send the list of players and their scores to a friend as an attachment.

State the precautions that need to be taken when sending and receiving the attachment.

.....

.....

.....

.....
.....
.....[3]

10.2 Effective use of the internet

S25_QP13_Q11

Alexander is considering buying a new smart TV and is carrying out research on the internet.

(a) Discuss how Alexander can evaluate the reliability of information found on the internet.

[6]

M25_QP12_Q12

Parents are worried about their children's use of the internet.

(a) Explain how parents could monitor the use of the internet by their children.

.....[3]

11 File management

11.1 Manage files effectively

S25_QP11_Q5

Viewers can watch movies by video streaming. One of the issues with video streaming can be a reduction in quality due to file compression.

(b) Explain what is meant by file compression.

.....

.....

.....

.....

.....

.....[3]

S25_QP11_Q14

Give **one** example of a generic file extension that would be part of the following types of file.

(a) image

.....[1]

(b) text

.....[1]

(c) web page

.....[1]

M25_QP12_Q13

(a) State the need for generic file formats.

.....

.....

.....

.....[2]

(b) Give two examples of generic file formats used for images.

.....

.....[2]

11.2 Reduce file sizes for storage or transmission

S25_QP13_Q11

Alexander is considering buying a new smart TV and is carrying out research on the internet.

- (b) Alexander has written a report which includes images and text. He was planning to store this document on his pendrive and finds that the document is too large to store.

State one way in which he could reduce the size of the document so it can be stored without removing any of the text or images.

.....
.....[1]

12 Images

12.1 Images

13 Layout

13.1 Create or edit a document

13.2 Tables

13.3 Headers and footers

14 Styles

14.1 Styles

S25_QP11_Q7

Explain, giving examples, the purpose of a corporate house style.

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.....[4]

15 Proofing

15.1 Software tools

S25_QP11_Q11

A student is typing a report for his biology coursework and needs to check his spelling for errors. The student could use spell check software.

(a) **Explain** how spell check software works.

.....

.....

.....

.....[2]

(b) Sometimes the spell check software highlights words or changes words which are correctly spelt.

Explain why this could have occurred.

.....

.....

.....

.....[2]

15.2 Proofing techniques

16 Graphs and charts

16.1 Graphs and charts

17 Document production

17.1 Document production

S25_QP13_Q11

Alexander is considering buying a new smart TV and is carrying out research on the internet.

(c) Alexander has added section breaks and a gutter margin to his report

Explain the purpose of:

(i) a section break

.....

.....

.....

.....[2]

(ii) a gutter margin

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.....

.....[2]

18 Databases

18.1 Create a database structure

S25_QP13_Q12

Describe the differences between a flat file database and a relational database.

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.....[4]

18.2 Manipulate data

18.3 Present data

19 Presentations

19.1 Presentations

20 Spreadsheets

20.1 Create a data model

S25_QP13_Q13

A spreadsheet uses both functions and formulas.

(a) **Explain** the differences between a function and a formula.

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.....[3]

(b) **State** one example of a:

(i) Function

.....

.....[1]

(ii) Formula

.....

.....[1]

(c) A spreadsheet uses relative cell referencing.

(i) **Explain** what is meant by relative cell referencing.

.....

.....[1]

(ii) **Describe** a problem that could occur when a relative cell reference is used.

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.....[2]

20.2 Manipulate data

20.3 Present data

21 Website authoring

21.1 Web development layers

W25_QP12_Q1

Circle **two** web development layers which are used when creating a web page.

Behaviour	CSS	Development	HTML
Operational	Presentation	Programming	System

[2]

S25_QP13_Q14

State two web development layers.

.....

.....

.....

.....[2]

21.2 Create a web page

W25_QP11_Q11

A student is creating a web page using HTML. She has coded the head section.

State the purpose of the head section. Include in your answer one element found in the head section

.....

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.....[2]

21.3 Use stylesheets

W25_QP12_Q2

A computer system uses both an analogue to digital converter (ADC) and a digital to analogue converter (DAC).

(a)

(i) Apart from converting data, **state** one other reason for using an ADC.

.....[1]

(ii) Apart from converting data, **state** one other reason for using DAC.

.....[1]

(b) **State** one device that could be used to input data into an ADC.

.....[1]

(c) **State** one device that could use the data output from a DAC

.....[1]

W25_QP12_Q3

A student is planning to buy a new desktop computer. He could buy one with a touch screen or one with a keyboard and mouse. **Discuss** the advantages and disadvantages of buying a desktop computer with a touch screen rather than one with a keyboard and mouse.

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.....[6]

W25_QP12_Q4

Compare internal memory and backing storage. Include in your answer both similarities and differences.

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.....[4]

W25_QP12_Q5

Tawara Cars manufactures and sells four models of a sports car. It uses a paper-based data capture form to record details about these models. It is planning to set up an online data capture form. The current form is shown.

Registration Number:	
----------------------	--

Please tick the Model of car

Model of car:	L-sport	E-sport	G-sport	H-sport
Date of registration:				
Sale price	\$			

Please tick the State of car and the Type of car

State of car:	New	Used		
Type of car:	Electric	Petrol	LPG	Hydrogen

(a) **State**, giving reasons for your choice, the most appropriate field name that could be used with the following design features. Your field names must be different in each case.

(i) Drop down list

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.....

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.....[2]

(ii) Radio button

.....
.....
.....[2]

(iii) Input Mask

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.....
.....[2]

(b) A robot paints cars during manufacture. The robot can be programmed by a set of instructions typed into the computer system.

Explain one other way in which the robot could be programmed to carry out this task.

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.....[2]

(c) **Discuss** the benefits and drawbacks of using a robot to paint the cars.

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.....[6]

W25_QP12_Q6

Students can bring their mobile computers into school to complete their work in lessons.

State two examples of mobile computers

[2]

W25_QP12_Q7

The owner of a theme park has decided to redesign the tickets for its rides. She is planning to use either a bar code or an RFID tag on the ticket.

(a) Describe the advantages of using an RFID tag rather than a bar code.

[6]

(b) The tickets for the theme park are booked online.
Describe the characteristics of an online booking system.

.....
.....[4]

W25_QP12_Q8

The owner of a theme park has decided to redesign the tickets for its rides. She is planning to use either a bar code or an RFID tag on the ticket.

(a) Describe the advantages of using an RFID tag rather than a bar code.

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.....[4]

(b) State two ways in which a user can attempt to prevent pharming attacks.

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.....[2]

W25_QP12_Q9

A student is creating a web page using HTML. The student has coded the body section.

State the purpose of the body section. Include in your answer two elements that could be found in the body section.

Purpose.....
.....

Element 1

Element 2[3]

W25_QP12_Q10

A network manager stores the backup of a school network on magnetic tape.

- (a) **Describe** the benefits and drawbacks of using magnetic tape to store backups

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.....[4]

- (b) The music teacher wants to use Bluetooth headphones on the computers in her music classes. **Explain** how the network manager could set up the Bluetooth headphones.

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.....[4]

W25_QP12_Q11

Confidential emails need to be secure in order to protect the privacy and confidentiality of the data.

- (a) **Describe** how this could be achieved using encryption of the email.

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.....[4]

(b) **State** two methods of data transfer that use encryption.

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.....[2]

W25_QP12_Q12

A family has purchased smart CCTV cameras to protect their home when they are at work.
These cameras can be monitored from a smartphone.

(a) **Describe** the positive and negative effects of using smart CCTV cameras.

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.....[5]

(b) **Identify** two other devices in the home that could be microprocessor controlled.

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.....[2]

W25_QP12_Q13

3D printing is used in medicine to create prosthetics.

(a) **State** two other uses of 3D printing in medicine.

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.....[2]

(b) State two benefits and two drawbacks of using 3D printing when creating prosthetics.

Benefit 1

Benefit 2

Drawback 1

Drawback 2[4]

W25_QP12_Q14

A school crossing is controlled by traffic lights. When pupils wish to cross the road, they press the button on the crossing control.

Identify two output devices that could be used in the crossing control.

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.....[2]

W25_QP13_Q1

Circle two direct data entry devices.

Actuator	Keyboard	Mouse	OMR	
Plotter	QR scanner	ROM	Sensor	[2]

W25_QP13_Q2

A computer system uses random access memory (RAM). **State** four characteristics of RAM.

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.....[4]

W25_QP13_Q3

A company delivers meals to elderly people. The delivery drivers use both an expert system and satnav to plan their deliveries.

(a) **Describe** how an expert system can be used to produce a schedule for the delivery of the meals.

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.....[6]

- (b) When the schedule has been produced, the delivery driver uses satnav to find the shortest route to the first person's address.

Describe how satnav can be used to find the shortest route to the address.

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.....[6]

W25_QP13_Q4

Voice recognition is used to allow a student to log into their laptop computer.

- (a) **Discuss** the impact of using voice recognition in this way.

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.....[6]

- (b) Voice recognition is a form of biometric password.

State two other forms of biometric password that could be used in this way.

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.....[2]

- (c) The student is planning to purchase a laptop computer which uses zero login authentication. **Discuss** the benefits and drawbacks of using zero login authentication.

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.....[4]

W25_QP13_Q5

Tawara Cars sells sports cars.

Its cars are either electric, LPG or hydrogen powered.

It sells three different models of its cars.

Cars being sold are either new or used.

The owner of Tawara Cars has set up a database to store the details of these cars. Part of the database is shown.

Reg_number	Model	Date_of_reg	Sale_price	State_of_car	Engine_type
VX231AA	Premier	10/03/2024	\$42000.00	U	LPG
AB764CJ	HydroSport	11/04/2025	\$126000.00	N	Hydrogen
FF269DA	EcoSport	15/01/2024	\$68000.00	U	Electric

- (a) **State**, giving one reason for your choice, the most appropriate validation check on data entered into each of the following fields.

- (i) Date_of_reg

.....
.....[2]

- (ii) Sale_price

.....
.....[2]

(iii) State_of_car

.....
.....[2]

(b) The owner of Tawara Cars is planning to code the model field to save space in the database.

(i) **State** a suitable code he could use.

.....[1]

(ii) Saving space is one reason for coding a field.

State three other reasons for coding this field.

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.....[3]

W25_QP13_Q6

When planning a new computer system, questionnaires or interviews can be used during the analysis stage.

(a) **Discuss** the advantages and disadvantages of using questionnaires rather than interviews.

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.....[6]

(b) A system specification is part of the analysis of a system.

Explain what is meant by a system specification. Include in your answer one item that is part of the system specification.

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.....[2]

W25_QP13_Q7

An examination board needs to hold a meeting with its examiners. They have decided to hold a video conference.

(a) State three items of hardware that would be needed to hold the video conference.

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.....[3]

(b) Describe the benefits to the examination board of holding a video conference.

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(c) Describe one piece of software that is required for the video conference.

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.....[3]

W25_QP13_Q8

A 3D printer is a form of inkjet printer. **Contrast** a 3D printer with an inkjet printer.

W25_QP13_Q9

Smishing can be a threat to data that is stored on computer systems.

(a) **Explain** what is meant by smishing.

[4]

[4]

(b) **State** two ways in which a user can attempt to prevent smishing attacks.

[2]

W25_QP13_Q10

A student is creating a web page using HTML. The web page has two sections.

(a) **State** the two sections found in a web page.

[2]

(b) The student is planning to place a bookmark for this web page.

Explain the concept of a bookmark.

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.....[2]

W25_QP13_Q11

Blogs and wikis are used on the internet.

(a) State two characteristics of a blog.

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.....[2]

(b) State two characteristics of a wiki.

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.....[2]

W25_QP13_Q12

A school network manager backs up work files onto a magnetic tape. The headteacher of the school thinks this is old technology and has asked the network manager to backup work onto a portable solid state drive (SSD).

Describe the advantages and disadvantages of using a magnetic tape to store backups rather than a portable SSD.

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.....[4]